

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Electronics & telecommunication Semester :M Subject Code & Name:BTETC601 Antennas & wave propogation Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
			Marks
Q. 1	Solve Any Two of the following.		
A)	Sketch the structure of atmosphere & explain each layer, significance of all		6
B)	Define the terms virtual height, MUF & their relevance in wave propagation		6
C)	Draw and explain multi hop propagation system		6
Q.2	Solve Any Two of the following.		
A)	What is fading effect? What are the different types of fading?		6
B)	Discuss the analysis of a small dipole antenna		6
C)	Derive radiation power density		6
Q. 3	Solve Any One of the following.		
A)	Discuss antenna parameters		12
B)	Calculate the length of dipole ,reflector, director 1, director 2 ,director 3 along with the spacing between element for channel 5 VHF = III the frequency band for Cj channel 5 is 174 MHZ to 181 MHZ		12
Q.4	Solve Any Two of the following.		
A)	Explain broadside array concept		6
B)	Compare broadside array & end fire array		6
Q. 5	Solve Any two of the following.		
A)	Draw & explain V antenna		6
B)	Draw & explain helical antenna		6
C)	Explain yagi uda antenna		6

	*** End ***	
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